PRISMA 2020 Eligibility Stage: Quality Criteria Scoring — Section 3.1 (Theoretical Foundations of Distributional Change)

Report prepared 10 August 2026

1. Scope of This Scoring Pass

This report extends the Step 4 eligibility scoring to the full candidate set retrieved for Section 3.1 (Theoretical Foundations of Distributional Change: The Dataset Shift Taxonomy) of the review, comprising all 14 full-text articles retained for this subsection following Step 3 screening. Each article was scored honestly against its full text rather than against title and abstract alone.

QC1 (problem statement and scope clarity) was scored against fit to this review's specific research question, namely concept drift, spatial domain shift, and continual or federated adaptation in machine learning-based IoT intrusion detection, appraised through the covariate-shift, prior-probability-shift, and concept-shift taxonomy established in Section III.A, rather than against the clarity of the problem statement in the article's own, different scope. An article can present an impeccably clear problem statement for a different question, such as domain-generalisation algorithm design, noise-robust domain adaptation, or graph neural network generalisation-bound derivation, and still score 0 on QC1 here, because that problem statement is not the one this review is screening for, and none of the three non-manuscript-cited articles engages the taxonomy as a diagnostic or appraisal instrument in the way the eleven manuscript-cited articles do.

QC4 (empirical extraction detail) was scored as 0 wherever an article's full text contained no drift-type, distributional-component, or adaptation-mechanism data extractable into this review's own extraction fields, regardless of how much extraction detail the article offered for its own, different research question. This reflects a consistent, review-specific application of the criterion rather than a per-article adjustment.

QC2 (methodological transparency) and QC3 (venue indexing and citability) were scored on their ordinary, article-intrinsic basis, since these criteria measure the rigor and citability of the work itself rather than its fit to this particular review.

2. Quality Assessment Criteria Applied

Table 1 reports the four quality criteria (QC1 to QC4) applied in this eligibility round, together with the scoring scheme used for each. A quality score is computed as the sum of applicable criterion scores divided by the number of applicable criteria, scaled to a 0 to 4 range.

Table 1. Quality assessment criteria applied to Section 3.1 candidate articles

Code	Criterion	Scoring
QC1	Problem statement and scope clarity	Yes (1) / Partial (0.5) / No (0)
QC2	Methodological transparency	Yes (1) / Partial (0.5) / No (0)
QC3	Venue indexing and citability	Yes (1) / Partial (0.5) / No (0)
QC4	Empirical extraction detail (where applicable)	Yes (1) / Partial (0.5) / No (0) / N/A

Articles published in 2020 or later were retained where their total quality score met or exceeded 2.5 out of 4. Articles published before 2020 were retained where their total quality score met or exceeded 1.5 out of 4.

3. Results

Of the 14 full-text articles assessed for Section 3.1 against the quality criteria in Section 2, 11 articles met or exceeded their applicable threshold and were retained, while 3 articles fell below their applicable threshold and were excluded. Table 2 lists each assessed article together with its per-criterion and total quality scores, applicable threshold, and eligibility decision, sorted with included articles listed first, followed by excluded articles.

Table 2. Quality criteria scoring results for Section 3.1 candidate articles, sorted by decision

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
1	A Survey on Transfer Learning (2010)	2010	1.0	1.0	1.0	1.0	**4.0**	≥1.5	**Include**
2	Improving predictive inference under covariate shift by weighting the log-likelihood function (2000)	2000	1.0	1.0	1.0	1.0	**4.0**	≥1.5	**Include**
3	A unifying view on dataset shift in classification (2012)	2012	1.0	1.0	1.0	1.0	**4.0**	≥1.5	**Include**
4	A unified framework for dataset shift diagnostics (2023)	2023	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
5	Efficient and Multiply Robust Risk Estimation under General Forms of Dataset Shift (2024)	2024	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
6	A Survey of Unsupervised Deep Domain Adaptation (2020)	2020	1.0	1.0	1.0	0.5	**3.5**	≥2.5	**Include**
7	A Comprehensive Survey on Test-Time Adaptation Under Distribution Shifts (2025)	2025	1.0	1.0	1.0	0.5	**3.5**	≥2.5	**Include**
8	Evaluation of domain generalization and adaptation on improving model robustness to temporal dataset shift in clinical medicine (2022)	2022	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
9	When Invariant Representation Learning Meets Label Shift: Insufficiency and Theoretical Insights (2024)	2024	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
10	Preventing dataset shift from breaking machine-learning biomarkers (2021)	2021	1.0	0.5	1.0	0.5	**3.0**	≥2.5	**Include**
11	Handling Out-of-Distribution Data: A Survey (2025)	2025	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
12	On the value of label and semantic information in domain generalization (2023)	2023	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
13	Towards Accurate and Robust Domain Adaptation Under Multiple Noisy Environments (2023)	2023	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
14	Generalization Error Bounds for Graph Neural Networks in Unseen Domains (2025)	2025	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**

This retained set of 11 articles constitutes the eligible pool for Section 3.1 (Theoretical Foundations of Distributional Change) and forms part of the input to the synthesis and thematic analysis stage.

4. Data Integrity Notes

Each article was scored against the four quality criteria on the basis of its full text, applying the review-specific interpretation of QC1 and QC4 set out in Section 1 above. The retain or exclude decision was applied strictly according to the threshold in effect for that article's publication year, namely 2.5 out of 4 for articles published 2020 or later, or 1.5 out of 4 for articles published before 2020. No article was retained or excluded on any basis other than its quality score relative to its applicable threshold.

5. Document Preparation

This report has been prepared in accordance with the quality assurance procedure prior to proceeding to the synthesis stage.

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